



UNDERGRADUATE PROGRAMMES

MAINSTREAM

(Regular/Morning & Evening Sessions)

END OF SECOND SEMESTER IN-PERSON EXAMINATIONS - 2020/2021 ACADEMIC YEAR

FACULTY OF ACCOUNTING AND FINANCE

DEPARTMENT OF BANKING AND FINANCE

LEVEL 200

BCPC 208: QUANTITATIVE METHODS

Date: THURSDAY, AUGUST 5, 2021

TIME ALLOWED: 3 HOURS

INDEX NUMBER

INSTRUCTION(S):

ATTEMPT ANY THREE (3) QUESTIONS

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Undergraduate Library

Question 1

The Coca-Cola Company is an American multinational corporation, and manufacturer, retailer, and marketer of non-alcoholic beverage concentrates and syrups. The company is best known for its flagship product Coca-Cola, invented in 1886 by pharmacist John Stith Pemberton in Atlanta, Georgia. In Ghana, the company was formed from a divestiture of Bottling Division of G.N.T.C in March 1995 and started operations on March 7, 1995, at the GNTC plant at Adjabeng. The new premises sited off the Spintex Road, was commissioned in 1996. Since the divestiture, the Company has invested over US\$90 million in vehicles, glass bottles, plastic bottles, plastic crates, production, and marketing equipment. In 2020, the company contracted an economist who estimated the demand function for the company's product (Coca-Cola) in the two biggest cities in Ghana, Accra and Kumasi as follows:

$$\text{Kumasi: } 1.5Q_1 = 50 - P_1 + Q_2$$

$$\text{Accra: } 1.5Q_2 = 52.5 + Q_1 - P_2$$

In addition, Coca Cola Bottling company has a joint total cost function as:

$$TC = 5Q + 4Q_1Q_2$$

where $Q = Q_1 + Q_2$ and Q_1 and Q_2 denote the quantities of Coca Cola demanded in Kumasi and Accra, respectively

- (a) Find the Total Revenue function **6 marks**
- (b) Find the profit function for the firm. **2 marks**
- (c) Determine the output level that maximizes the firm's profit using the first and second order conditions. **12 marks**

Question 2

Maria has an online shop where she sells handmade paintings and cards. She sells the painting for €70 and the card for €90. It takes her 1 and half hours to complete a painting and 3 hours to make a single card. She also has a day job and makes paintings and cards in her free time. She cannot spend more than 135 hours a month to make paintings and cards. Packing of the products and delivering it to the final customer cost Maria 2 hours of packaging time for paintings and an hour for cards. In addition, she cannot spend more than 100 hours a month on packaging these products. Additionally, the total output of paintings and cards that Maria makes a month should not be more than 60. In all, it costs Maria €60 and €40 to produce and deliver the paintings and cards, respectively to the final customer.

- (a) Formulate a linear programming model for the above managerial problem. **3 marks**
- (b) How many paintings and cards should Maria produce to maximise profit. **9 marks**
- (c) Find the maximum profit. **1 mark**
- (d) Determine the scarce resources **1 mark**
- (e) Find the shadow price of each of the scarce resource. **6 marks**

Question 3 (a)

AFB Company Ltd is a company that was set up in June 2020 that produces a fertilizer known BedRock Fertilizer. Among the competitors of your product are Asaase Wura fertilizer from YARA Ghana Ltd. As an expert in the marketing of fertilizers, you have been poached from YARA Ghana Ltd to oversee the marketing of this new product. For you to make informed decision and advise the board appropriately, you engaged an economist who estimated the demand equation for your firm's product. The economist used data from 549 cocoa farmers around the country for the month of January 2021 to estimate a demand curve for your product as:

$$Q_d + 0.25P - 12 = 0$$

In addition, the economist also estimated the supply function for your product as:

$$4Q_s - 2P = 6$$

- (i) Determine the equilibrium level of prices and quantity demand. **4 marks**
- (ii) Represent your answer graphically **4 marks**
- (iii) Determine the welfare of economic agents. **2 marks**
- (iv) Suppose the government imposes a tax of €6 on every bag sold, determine the new equilibrium price and quantity. **3 marks**
- (v) Show how the economic agents share the tax. **2 marks**
- (vi) What is the amount of government tax revenue? **1 marks**

Question 3 (b)

If fixed costs are 12, average variable cost are $(Q + 3)$, and the inverse demand curve is

$$P = 14 - Q$$

- (i) Obtain an expression for profit in terms of quantities **1 mark**
- (ii) For what values of Q does the firm breakeven? **3 marks**

Question 4

You have been appointed the new manager for Ghana Airways Company Limited, an international airline company that flies from the Kotoka International Airport in Accra to Heathrow Airport in London every day. The airline is described as a monopolist and has the possibility of discriminating between its Business and Economy Travelers. To help you price your services appropriately to maximize profit, you engaged an economist who estimated the demand function for both Economy and Business Travelers as:

$$\text{Economy Class: } P_1 = 40 - Q_1 - Q_2$$

$$\text{Business Class: } P_2 = 100 - Q_1 - 4Q_2$$

Where Q_1 and Q_2 are the respective numbers of Economy and Business Travelers and P_1 and P_2 are their respective fares. If the total cost of this airline company for flying these two categories of travellers is given as:

$$TC = 5Q_1 + 10Q_2$$

- (a) Find the profit function for the firm. 4 marks
- (b) Determine the firm's pricing policy that maximizes profit if the firm's total costs are fixed at €100. 12 marks
- (c) Find the maximum profit of the firm. 1 mark
- (d) Estimate the new optimal profit if the fixed costs rise to €103 3 marks

Question 5 (a)

A monopolist's demand function is

$$0.25Q = 12.5 - 0.5P$$

The fixed costs of production are 7 and the variable costs are $(Q + 1)$ per unit.

- (a) Find the value of output that maximizes profit 4 marks
- (b) Find the maximum profit 1 mark

Question 5 (b)

At the beginning of 2020, January 1, three sisters Ama, Efua and Baaba were given an amount of ₦10,000 to be shared and invested in an interest-bearing account. Ama was given an amount x , with which she was able to negotiate an interest of 5% per annum. Efua was, also, given an amount y , which he invested in an account that pays an interest of 10% per annum. Baaba, on the other hand, was given an amount z , which he invested in an account that earned her an interest of 8% per annum on his investment. At the end of the year, the three sisters were able to earn a total interest of ₦800. In addition, supposed the amount of money given to Baaba is ₦400 less than the amount of money given to Ama and Efua combined.

- (i) Write down the three linear equations **2 marks**
- (ii) Represent the system of equations in a matrix form $AX = B$ **1 mark**
- (iii) Find the determinant of the matrix A **2 marks**
- (iv) Find the inverse of the matrix A **8 marks**
- (v) Find the amount of money given to each of the 3 siblings using matrix inversion **2 marks**

END OF PAPER

Examiner: Ebo Botchway, (PhD)